

Grade 12 Chemistry Unit 3 Industrial Chemistry

3.1 Introduction

At the end of this section, you will be able to:

- define industrial chemistry
- list general characteristics of chemical industries
- mention the role of industrial chemistry in the society

3.2 Natural Resources and Industry

At the end of this section, you will be able to:

- define natural resources
- list the types of natural resources
- classify chemical industries based on the raw materials they use to manufacture their products
- classify chemical industries based on the type of product they manufactured
- describe the applications of minerals in industry.

3.3 Manufacturing of Valuable Products/ Chemicals

At the end of this section, you will be able to

- describe the general methods of “manufacturing of valuable products in industries”
- explain the important steps for the production of NH_3 in Haber process
- use Lechaterier’s principles to explain how the yield of ammonia can be increased in Haber process
- consult the internet or other reference materials to inspect some important direct uses of H_2SO_4 , HNO_3 and Na_2CO_3
- examine the physical and chemical properties of H_2SO_4 , HNO_3 , and Na_2CO_3

- describe the important steps for the production of HNO_3 in Ostwald Process
- list the physical and chemical properties of HNO_3
- explain how the production of H_2SO_4 is considered as the vital to measure the development of a Nation
- draw the schematic diagram which shows the steps followed in the production of H_2SO_4 in contact process
- state the raw material used to manufacture Na_2CO_3
- draw the schematic diagram which shows the steps followed in production of Na_2CO_3 in Solvay process
- discuss the uses of DAP, $(\text{NH}_4)_2\text{HPO}_4$, Nitrogen fertilizers, herbicides and insecticides in the agriculture sector
- apply indigenous knowledge to produce natural compost
- use local resources to prepare insecticides.

3.4 Some Manufacturing Industries in Ethiopia

At the end of this section, you will be able to

- list some manufacturing industries in Ethiopia
- outline the important steps in the production of ceramics
- mention some uses of ceramics
- Write the important steps in the production of cement, ceramics, glass, sugar and ethanol
- predict the product of sugar fermentation
- prepare ethanol from locally available ingredients compare and contrast the indigenous and industrial methods of: food preservation and packing, making ethanol, and processing skin and hide.
- compare and contrast locally made clay materials with ceramics
- outline the important steps in the production of pulp and paper, soaps and detergent and dry detergents

- **explain how tanning is carried out**
- **mention some uses of skin and hides**
- **present a report to the class after visiting a nearby food factory.**